$L[\sin\omega t]=\frac{\omega}{s^2+\omega^2}$ $L[t^2]=\frac{\Gamma(2+1)}{s^{2+1}}$ $L(e^{2t})=\frac{1}{s-2}$ $L[u(t-1)]=\frac{1}{s}e^{-s}$

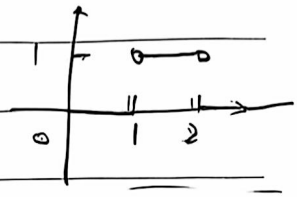
第七次作业补

习题七：1(1)(4) 2(2) 3(3) 5(5)(6) 6(2) 8(3) 9(6)

T1 (1) $L[f(t)]=L[\frac{1}{2}\sin 2t]=\frac{2}{2(s^2+4)}=\frac{1}{s^2+4}$

(4) $L[f(t)]=L[t^2]=\frac{\Gamma(3)}{s^3}=\frac{2}{s^3}$

T2 (2) $L[f(t)]=\int_0^\infty f(t)e^{-st}dt=-\frac{1}{s}(e^{-2s}-e^{-s})$



T3 (3) $F(s)=\frac{1}{s^2}-\frac{1}{s^3}+\frac{1}{s^4}-\frac{1}{s^5}$

$L^{-1}[F(s)]=t-\frac{1}{2}t^2+\frac{1}{6}t^3-\frac{1}{24}t^4$

T5 (5) $f(t)=[(t-1)^2+2(t-1)+1]u(t-1)$

$L[f(t)]=L[(t-1)^2u(t-1)]+2L[(t-1)u(t-1)]+L[u(t-1)]$

$=e^{-s}\{L[t^2]+2L[t]+L[1]\}$

$=e^{-s}(\frac{2}{s^3}+\frac{2}{s^2}+\frac{1}{s})$

(6) $L[t^{\frac{1}{2}}]=\frac{\frac{1}{2}\Gamma(\frac{1}{2})}{s^{\frac{3}{2}}}$ 故 $L[t^{\frac{1}{2}}e^{\lambda t}]=\frac{1}{2\sqrt{2}(s+\lambda)^{\frac{3}{2}}}$

T6 (2) 设 $f_1(t)=\begin{cases}1 & 0<t<1\\ 0 & \text{其他}\end{cases}$，则 $L[f_1(t)]=\int_0^1 e^{-st}dt=\frac{1}{s}(1-e^{-s})$

$f(t)=f_1(t)+f_1(t-2)+f_1(t-4)+\cdots=$

故 $L[f(t)]=L[f_1(t)](1+e^{-2s}+e^{-4s}+\cdots)=\frac{1-e^{-s}}{s(1-e^{-2s})}=\frac{1}{s(1+e^{-s})}$

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T8 (3) $L\left[\int_0^t \frac{e^{-x}\sin x}{x}dx\right] = \frac{1}{s}L\left[\frac{e^{-t}\sin t}{t}\right]$

$L\left[\frac{\sin t}{t}\right] = \int_s^\infty \frac{1}{s^2+1}ds = \operatorname{arctan} s\Big|_s^\infty = \frac{\pi}{2} - \operatorname{arctan} s$

故原式 $= \frac{1}{s}\left[\frac{\pi}{2} - \operatorname{arctan}(s+1)\right]$

T9 (6) $-F'(s) = \frac{2}{s(s^2+1)} = \frac{2}{s} - \frac{2s}{s^2+1}$

$F(\infty)=0$, $L[2-2\cos t] = -F(s)$ 故 $L\left[\frac{2-2\cos t}{t}\right] = F(s)$

即 $L^{-1}[F(s)] = \frac{2-2\cos t}{t}$

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